STEP 6 OF 8

OBC and Flight Software

On-Board Computer and OBSW — FreeRTOS · ARM Cortex-M4 · MTQ-Only Revision

ARM Cortex-M4F | FreeRTOS | 7 RTOS Tasks | 30 Telecommands | 4-Level FDIR | RW Task Removed | SGP4 Orbit Propagator

6

OFFICIAL MISSION PARTNER CO-DESIGN DOCUMENT

MTQ-Only Revision | Reaction Wheel Removed | 3-Axis Magnetorquer System

1 OBC Overview — The Brain of SLARS-BCG-1

The On-Board Computer (OBC) is the central intelligence of SLARS-BCG-1. Every autonomous decision — when to image, when to downlink, how to respond to a fault, how to manage battery state — is executed by the OBC running the On-Board Software (OBSW). The OBC has no ground contact for ~86 minutes between passes and must operate independently for the entire mission lifetime.

ARM M4	FreeRTOS	7 Tasks	512 MB	30 TCs
Processor @ 120 MHz	Real-time OS	RTOS tasks (was 8)	eMMC science flash	Command set (was 32)

MTQ-ONLY OBSW CHANGES — SUMMARY

1. FreeRTOS tasks: 8 -> 7 (RW monitor/desaturation task removed entirely)
2. Science window gate: attitude error < 5 deg -> < 8 deg (MTQ accuracy)
3. TC 0x11 SET_RW_SPEED: removed | TC 0x14 TRIGGER_DESATURATION: removed
4. TM housekeeping field rw_speed: removed from beacon frame
5. FDIR: RW saturation check (> 5,500 RPM) removed from monitor task
6. IGRF-13 model retained — still used for MTQ PID B-field direction reference

2 OBC Hardware Specification

Parameter	Specification	Rationale
Processor	ARM Cortex-M4F @ 120 MHz	FPU essential for EKF and SGP4 float arithmetic
RAM (SRAM)	256 KB internal + 2 MB external	Task stacks + APRS buffer + image frame buffer
Program flash	1 MB internal NOR flash	OBSW binary fits with comfortable margin
Science data storage	512 MB eMMC (industrial grade)	Buffers ~3 days of science data at full rate
IGRF model storage	512 KB dedicated NOR sector	MTQ control law B-field direction reference
Interfaces	I2C x3, SPI x2, UART x4, GPIO x16, ADC x8	All subsystems on standard embedded interfaces
Watchdog timer	Hardware WDT, 100 ms timeout	Independent of RTOS — fires even if kernel hangs
Real-time clock	Battery-backed RTC, +/-1 s/day	Mission time for pass prediction + TM timestamps
Radiation tolerance	Commercial grade + 2 mm Al shielding	0.9 krad/yr at 500 km — COTS acceptable
Power draw (active)	~0.5 W at 120 MHz	Within Step 2 OBC allocation
Power draw (sleep)	~0.05 W between task executions	CPU sleeps in WFI state
Recommended	GomSpace NanoMind A3200 or EnduroSat	OBC IIARM-based, PC/104, proven heritage
Custom option	STM32H743 + custom PCB	Higher performance, lower cost, student-buildable

3 OBSW Architecture — Layered Design

The OBSW follows a strict layered architecture — each layer depends only on the layer below it. This isolates hardware dependencies in the HAL, keeps the RTOS independent of application logic, and allows individual tasks to be tested and updated independently. Consistent with ECSS software engineering standard used by ESA.

Layer	Contents	Key Role
Hardware Abstraction (HAL)	Drivers: I2C, SPI, UART, GPIO, PWM, ADC	Isolates hardware from software
FreeRTOS Kernel	Scheduler, semaphores, queues, timers, ISR handlers	Real-time deterministic execution
Device Drivers	MTQ driver, radio driver, imager interface, AIS UART, EPS	I2CSubsystem abstraction
Core Services	TM manager, TC parser, flash filesystem, FDIR engine time service	Shared infrastructure
Application Tasks	Power manager, ADCS controller, payload scheduler, COMission logic	MS, APRS relay
Mission Management	Orbit propagator, pass predictor, science gating, EOL power reduction	Science intelligence
Ground Interface	TC dispatcher, TM frame builder, CFDP, CRC validation, HMAC auth	Command and telemetry

4 FreeRTOS Task Architecture — 7 Tasks (MTQ-Only Revision)

Removing the reaction wheel eliminates the need for a dedicated RW speed monitor and desaturation task. The OBSW runs 7 FreeRTOS tasks — down from 8. The ADCS controller task handles all three MTQ control modes. No momentum management is required.

Task Name	Period	Priority	Stack	Responsibility
Watchdog task	100 ms	8 — highest	512 B	Kicks hardware WDT; confirms all tasks alive within deadline
ADCS controller task	100 ms	7	4 KB	Sensor read, EKF update, B-dot/PD/PID MTQ control — no RW commands
FDIR monitor task	2,000 ms	6	2 KB	Checks all subsystem health; escalates fault Level 1 through 4
Power manager task	1,000 ms	6	2 KB	Reads EPS SoC; enforces mode transitions; TX sequencing rules
COMMS manager task	500 ms	5	4 KB	Builds TM frames; dispatches TCs; Doppler correction; CFDP
Payload scheduler task	1,000 ms	5	4 KB	SGP4 pass prediction; gates imager at < 8 deg; APRS queue management
Telemetry task	10,000 ms	4	2 KB	Assembles housekeeping TM beacon; logs to flash; updates TM store
Flash manager task	5,000 ms	3 — lowest	4 KB	eMMC wear levelling; LittleFS maintenance; old data purge
REMOVED: RW monitor	—	—	—	***Reaction wheel speed monitoring + desaturation — not needed***
Pass prediction horizon	24 hours ahead			Scheduler plans next day of science passes
Sri Lanka bounding box	Lat 5.9N-9.9N, Lon 79.7E-81.9E			2 deg margin around island boundary

5 Orbit Propagator — SGP4 for Pass Prediction

SGP4 Parameter	Value	Notes
Algorithm	SGP4 (AFSPC standard)	Industry standard for LEO orbit propagation
TLE format	Standard 2-line NORAD TLE	Uplinked via TC 0x02 from ground station
Accuracy (fresh TLE)	< 1 km position error	TLE age < 1 day
Accuracy (3-day TLE)	< 5 km position error	Still adequate for pass prediction +/-30 s

Pre-activation lead time	7 minutes before footprint entry	ADCS Mode 3 + imager warm-up time
TLE uplink frequency	At least every 3 days	Ground station uploads via TC uplink
Library	libsgp4 (C implementation)	Open source; < 50 KB; well-tested flight heritage
CPU load	~2% at 120 MHz	Runs in payload scheduler task every 60 s

6 Power Management Software

Rule	Trigger Condition	Action	Source
Safe mode entry	SoC < 20% OR voltage < 6.2 V	All payloads off; ADCS minimal; beacon only	Step 2
Safe mode exit	SoC > 70% AND safe for > 2 orbits	Return to nominal mode	Step 2
EOL science reduction	SoC < 60% at pass window start	Reduce science to 2 passes/day max	Step 2
Science mode gate	SoC > 60% AND ADCS error < 8 deg	Enable science mode for pass	Steps 2+5
TX sequencing	UHF PA commanded ON	Block S-band TX until UHF PA off	Step 4
Peak load guard	Total draw > 5.8 W	Disable lowest-priority payload	Step 2
Emergency power off	SoC < 10% OR thermal alert	Force-disable all non-essential loads	Step 2
REMOVED: RW desat gate	RW speed > 4,500 RPM	Not applicable — no reaction wheel	N/A

```
SCIENCE WINDOW GATE (updated for MTQ-Only):
  if att_err < 8.0 and soc > 0.60 and passes_today < max_passes:
      imager.enable()   # Power rail ON
      sdr.enable_tx()   # SDR relay TX ON
  # Previously: att_err < 5.0 (with reaction wheel assist)
  # Updated to: att_err < 8.0 (MTQ-only achievable accuracy)
```

7 Fault Detection, Isolation and Recovery — FDIR

FDIR Level	Trigger	Action	Recovery Time
Level 1 — Task reset	Task misses 3 WDT deadlines	FreeRTOS task_delete + task_create	< 1 second
Level 2 — Subsystem reset	Level 1 fails 3x on same subsystem	EPS rail off (10 s) then on	10–30 seconds
Level 3 — Safe mode	Level 2 fails OR SoC < 20% OR thermal	All payloads off; beacon only; charge	1–3 orbits
Level 4 — Emergency TC	Level 3 for > 3 orbits without recovery	Transmit anomaly report; await TC	Next ground contact

SLARS-BCG-1 implements a four-level escalating FDIR. Each level is tried three times before escalating. RW saturation check is removed — all other checks unchanged.

Battery SoC	20–100%	< 20% for 2 consecutive reads	Level 3
Battery voltage	6.0–8.4 V	< 6.0 V (under-voltage cutoff)	Level 3
OBC temperature	-20 to +70 deg C	< -25 or > +75 deg C	Level 2
ADCS angular rate	< 10 deg/s	> 15 deg/s for 30 s	Level 2
ADCS pointing error (Mode 3)	< 8 deg	> 14 deg for 60 s	Level 1
UHF COMMS beacon	Beacon confirmed	No beacon TX for 90 s	Level 1
eMMC flash health	Read/write OK	3 consecutive write failures	Level 2
EPS current total	< 5.5 W	> 5.8 W for > 5 s	Level 2
OBC watchdog	All tasks kick < 100 ms	Any task fails to kick	Level 1
REMOVED: RW speed	—	RW not present — check removed	N/A

8 Telecommand Set — 30 Defined Commands (MTQ-Only)

TC ID	Command	Parameters	Effect
0x01	SET_MODE	SAFE / NOMINAL / SCIENCE	Force system into specified operating mode
0x02	UPLINK_TLE	2-line TLE string (138 chars)	Update orbit propagator with fresh TLE
0x03	REBOOT_OBC	delay_s: 0–255	Scheduled OBC reboot after delay
0x04	RESET_SUBSYSTEM	subsystem_id: 0–7	EPS power rail off then on for subsystem
0x05	SET_MAX_PASSES	n: 1–6	Override daily science pass limit
0x06	FORCE_BEACON	—	Immediately transmit housekeeping TM frame
0x07	REQUEST_EXTENDED_TM	page: 0–7	Downlink extended telemetry page
0x08	ENABLE_PAYLOAD	payload_id: 0–2	Force enable imager / SDR / AIS
0x09	DISABLE_PAYLOAD	payload_id: 0–2	Force disable specified payload
0x0A	SET_ADCS_MODE	mode: 1 / 2 / 3	Override ADCS operating mode
0x0B	CLEAR_APRS_BUFFER	—	Flush all buffered APRS emergency messages
0x0C	START_SBAND_DL	file_id: 0–999	Begin S-band CFDP transfer of specific file
0x0D	FORMAT_FLASH	confirm: 0xDEAD	Emergency: erase and reformat science flash
0x0E	SET_DOPPLER_OFFSET	offset_hz: +/-100000	Manual Doppler frequency override
0x0F	UPDATE_IGRF	data: 512 KB block	Upload new IGRF-13 table (MTQ control reference)
0x10	SET_MTQ_DIPOLE	axis, value: A.m2	Direct MTQ dipole command on any axis
0x11	READ_FDIR_LOG	entries: 1–50	Download last N FDIR event log entries

Monitored Parameter	Normal Range	FDIR Trigger	Level

0x12	SET_BEACON_INTERVAL	seconds: 10–120	Change UHF beacon transmission period
0x13	SET_SL_BBOX	lat/lon min/max	Update Sri Lanka bounding box for pass detection
0x14–0x1F	RESERVED	—	Future expansion
REMOVED: SET_RW_SPEED—		***—***	***Reaction wheel removed — command deleted***
REMOVED: TRIGGER_DESAT—		***—***	***No RW to desaturate — command deleted***

STEP 6 LOCKED — 7 FreeRTOS tasks. 30 telecommands. Science gate < 8 deg. RW monitoring and desaturation removed. 4-level F

9 Step 6 Locked Output Parameters — Feeds Steps 7 and 8

Locked Parameter	Value (MTQ-Only)	Feeds Into
OBC processor	ARM Cortex-M4F @ 120 MHz with FPU	Step 7: mass 80 g, 0.25U, 0.5 W
RTOS	FreeRTOS v10.5+	Step 7: no licensing cost
Number of RTOS tasks	7 tasks (was 8 — RW task removed)	Step 7: memory allocation confirmed
TC command set	30 commands (was 32 — 2 RW TCs removed)	Step 8: ground station software
Science window gate	Attitude error < 8 deg (was 5 deg)	Step 8: GS monitoring threshold updated
FDIR: RW check	REMOVED — no reaction wheel present	Step 8: simplified anomaly response
Power sequencing	UHF PA and S-band TX never simultaneous	Step 7: confirmed — no structural change
FDIR escalation	4 levels: task reset, power cycle, safe, TC	Step 8: GS responds to Level 4
Boot time	< 7 seconds total (6 stages)	Step 8: GS times first contact accordingly
Software update	CFDP binary patch on S-band, CRC-gated	Step 8: GS CFDP uplink required
Development language	C (OBSW) + Python (ground tools)	Step 8: student team skills
OBC mass	~80 g	Step 7: mass budget item
OBC volume	0.25U (90x96x15 mm, PC/104)	Step 7: bus stack position
OBC average power	~0.5 W active, ~0.05 W sleep	Step 7: within Step 2 allocation